PROFESSIONAL PORTFOLIO

Lázaro Gómez Ontiveros

Data-driven service operations
Method, models, governance and applied AI cases

Chief Operations Officer · Technology services

Multi-country operations in LATAM · Telecommunications and managed services

Mexico City · lazarogomez@hotmail.com · +52 55 7980 3050

This document presents methods and frameworks developed throughout my career. All data, figures and scenarios are synthetic and were built to illustrate the method. It contains no information from any client, vendor or employer.

HOW TO READ THIS DOCUMENT

Eight pieces, one method

This portfolio does not showcase projects: it shows how I approach an operational problem. Each piece starts from a business question, applies an explicit method and ends in an artifact someone can use the next day. Pieces 7 and 8 take the same method into two complete data and artificial intelligence scenarios, with platforms, CAPEX and OPEX.

> *I turn service operations managed by perception into operations managed by data, without adding headcount.*

Contents

#	Piece	Question it answers	Page
1	Coverage sizing with hourly data	How many people do I need, at what hours, and can I extend service without hiring?	3
2	Operational level agreement framework	If I outsource the operation, how do I keep control and accountability?	5
3	Cooperation model between presales, engineering and operations	How do I avoid selling what operations cannot sustain?	7
4	Executive operations dashboard	What should leadership look at every month, and what not?	9
5	Service and deliverables matrix per contract	How does a commercial catalog become verifiable obligations?	11
6	Commercial intelligence on operational data	What should I offer each client, current or new, without committing what operations cannot sustain?	13
7	**Patient 360: data and AI in a private hospital** NEW	How do I anticipate what each patient needs, and what does it cost to build and run?	16
8	**Predictive operations in retail** NEW	How do I see a failure or a deviation before it hits sales, and what does it cost?	18
—	How I work	The method behind the eight pieces	20

Career at a glance

EXPERIENCE
20+
years in operations and technology

TELECOMMUNICATIONS
21
years in network operations and service delivery

SCOPE
LATAM
multi-country, multi-site operations

EDUCATION
MBA
ITAM (in progress) · B.Eng. Electronics, IPN · Certified Chief AI Officer

A note on the data

Every number in this portfolio is synthetic. The numbers keep the shape and typical behavior of a real operation —the hourly curve, the mix of priorities, the effect of alert bursts, the order of magnitude of a CAPEX— because that is what makes the method understandable, but they do not belong to any real organization. Platform prices are list prices and for reference only. Governance frameworks are presented as generic templates, not as documents from any contract.

01 Coverage sizing with hourly data

From the demand curve to the staffing decision

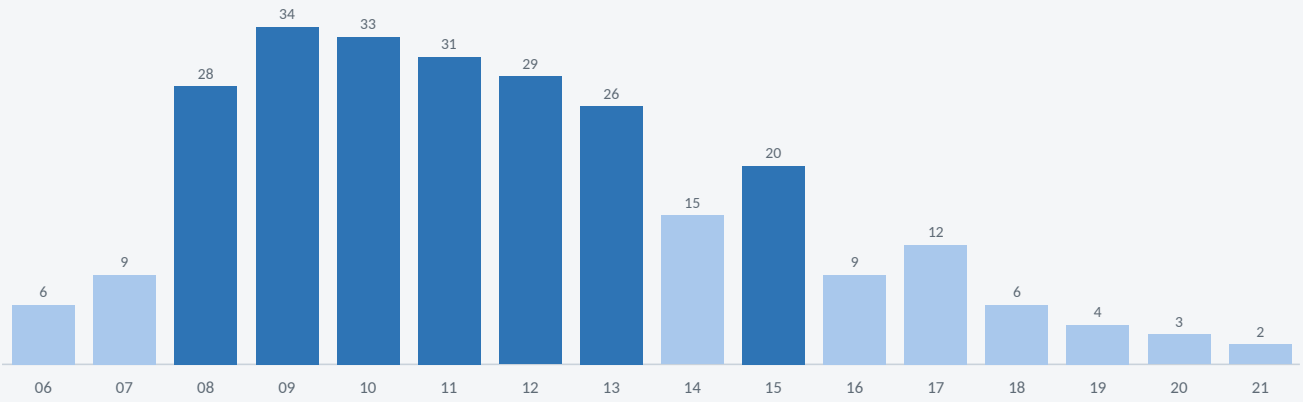
THE PROBLEM

A client asks to extend the service window from 9 to 14 hours a day. The intuitive answer is to hire more people and quote the increase. The right question is a different one: where does demand really sit within the day, and how much capacity is already being wasted in the hours when nothing happens?

THE METHOD

1	2	3	4	5
Extract record-level detail, not the monthly aggregate	Build the hourly demand curve and normalize bursts	Apply handling time and target occupancy	Calculate required resources per time slot with a coverage floor	Design shifts that cover the window with the existing team

STEP 1 · THE DEMAND CURVE (SYNTHETIC DATA, MONTHLY EVENTS BY HOUR)



Two immediate readings. Most demand sits between 08:00 and 13:00, not evenly spread across the day. And the 06:00–07:59 and 17:00–20:59 slots hold 16% of the events that nobody serves today: that is the case for extending the window, and there is no need to assume it.

STEP 2 · SEPARATING REAL DEMAND FROM NOISE

Automatic alerts triggered by a single root event —a link failure, a firmware update— arrive as dozens of records within minutes. If they are not grouped under a parent ticket, they distort three things at once: sizing, compliance measurement and the team's perception of workload. Normalization is the step that changes the result the most, and the one that is almost always skipped.

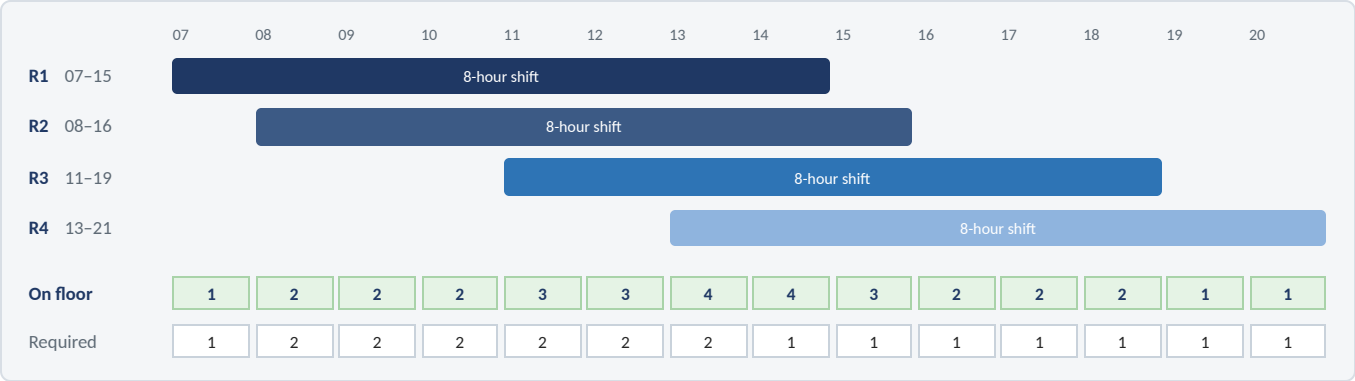
EVENTS IN THE MONTH	AFTER NORMALIZING	DIFFERENCE
338	**281**	**17%**
raw observed volume	real demand attributable to users	monitoring noise, not operational load

STEP 3 · FROM DEMAND TO RESOURCES PER TIME SLOT

The formula is deliberately simple, because it has to hold up in front of a client and in front of finance: resources per slot equal the hour's events divided by working days, times handling time, divided by target occupancy, with a floor of one resource in every hour of the window.

Assumption	Value	Why
Handling time	0.70 h	Observed median, not the mean: the mean gets distorted by long cases
Target occupancy	70%	Leaves 30% for meals, documentation, travel and variability
Working days	21	Monthly base for the period analyzed
Coverage floor	1 resource	Service continuity: no hour of the window is left unattended

STEP 4 · THE SHIFT PROPOSAL



Four staggered 8-hour shifts cover 07:00 to 21:00. The top row is the resulting staff on the floor; the bottom row is the calculated requirement. The mid-afternoon overlap is deliberate: it is the window for shift handover and staggered meals.

THE RESULT

SERVICE WINDOW

9 → 14 h

per working day

HEADCOUNT

No change

same people, redistributed

DEMAND COVERED

84 → 94%

+10 coverage points

The finding that changes the conversation

The total requirement of the extended window is 17 resource-hours a day, and the team on the floor provides 32. The problem was never a lack of people: the four people were all in the same time slot while 16% of demand happened outside of it. Extending hours was not a cost, it was a redistribution.

WHAT I WOULD DO DIFFERENTLY

I would measure actual touch time from the start instead of using median clock time. It is the only assumption with material weight on the result: if real touch time were double, the peak would rise to three resources and the conclusion would change. An honest model declares its fragile assumption.

02 Operational level agreement framework

What to sign when a third party runs the operation

THE PROBLEM

A company transfers its operation to a vendor and keeps contractual accountability toward its client. From that moment on, it answers for a service it does not run. Most agreements signed in this situation cover response times and little more, and leave out exactly what hurts later.

THE PRINCIPLE BEHIND THE WHOLE FRAMEWORK

> *The vendor's commitment must be stricter than the one the company signed with its client. The difference is the management margin.*

If both levels are identical, any vendor deviation immediately becomes a breach toward the client, with no room to validate, escalate or communicate. The practical rule: the internal target equals 80% of the contracted time, with an absolute margin floor for critical priority.

Item	Committed to client	Vendor target	Management margin
P1 resolution	4 h	3 h	1 h
P2 resolution	6 h	4 h 30 min	1 h 30 min
P3 resolution	8 h	6 h	2 h
Monthly availability	99.0%	99.5%	0.5 points

THE FIVE CHAPTERS ALMOST NOBODY INCLUDES

Chapter	What it prevents
Volume bands and demand management	That every load variation gets renegotiated without a rule. Defines base volume, tolerance band and marginal price outside the band.
Vendor operational continuity	That the agreement demands continuity of the client's service and says nothing about the operator's own platforms. Requires a plan, recovery times and an annual test.
Labor compliance and subcontracting	Joint liability when there is staff on the client's site. Periodic evidence and prior authorization of subcontractors.
Step-in right and assisted exit	That the highest level of consequences is a threat without a mechanism. Defines temporary step-in, causes and an exit fee agreed from the start.
Return of operational data	The silent loss of engineering capability: if the vendor operates, the vendor learns. Requires monthly record-level delivery in a usable format.

THE DELIVERABLES MATRIX, IN BOTH DIRECTIONS

An agreement that only lists the vendor's obligations is incomplete: half of all breaches originate in dependencies the contracting party did not deliver on time.

From vendor to contracting party

Deliverable	Frequency
Operations and compliance report	Monthly
Record-level data extract	Monthly
Root cause report	Per event
Reconciled inventory and configuration	Quarterly
Staffing and certifications status	Quarterly
Security incident notification	24 h / 72 h

From contracting party to vendor

Deliverable	Timing
Approved service design package	Before go-live
Tested runbooks	Before go-live
Access and credentials	Before go-live
Monitoring requirements and thresholds	Before go-live
Third-level support	On demand
Notice of scope change	30 to 60 days

RELIEF FOR AN UNMET DEPENDENCY

Exclusions do not work on their own. For the vendor to be relieved from the time count because of someone else's dependency, it must give notice within a defined period; without timely notice, the time counts against it. This is the clause that keeps blame from appearing for the first time at the monthly close, when it can no longer be verified.

ESCALATING CONSEQUENCES, IN BOTH DIRECTIONS

Level	Trigger	Consequence
1	One month below target	Documented action plan, with an owner and a date
2	Two consecutive months	Formal remediation plan with biweekly review
3	Three months or a passed-through penalty	Financial scheme and scope review
4	Repeated breach or serious failure	Step-in right and continuity assessment

And in the other direction: credit recovery after three months of sustained recovery, performance bonuses and shared savings from automation proposed by the vendor. An agreement that only punishes produces a vendor that argues about classifications instead of solving.

WHAT I WOULD DO DIFFERENTLY

I would negotiate the assisted exit fee at the beginning of the agreement, not at the end. It is the only moment when both parties have an incentive to be reasonable, and it is exactly the clause nobody wants to discuss when the relationship is just starting.

03 Cooperation between presales, engineering and operations

How to avoid selling what operations cannot sustain

THE PROBLEM

In many service organizations, the commercial design is done by presales and operations inherits the commitment. Presales optimizes to win; the cost of the mistake is paid by engineering, operations and the margin. The solution is not to put engineering into every opportunity —that turns it into a bottleneck— but to separate two things that today share the same name.

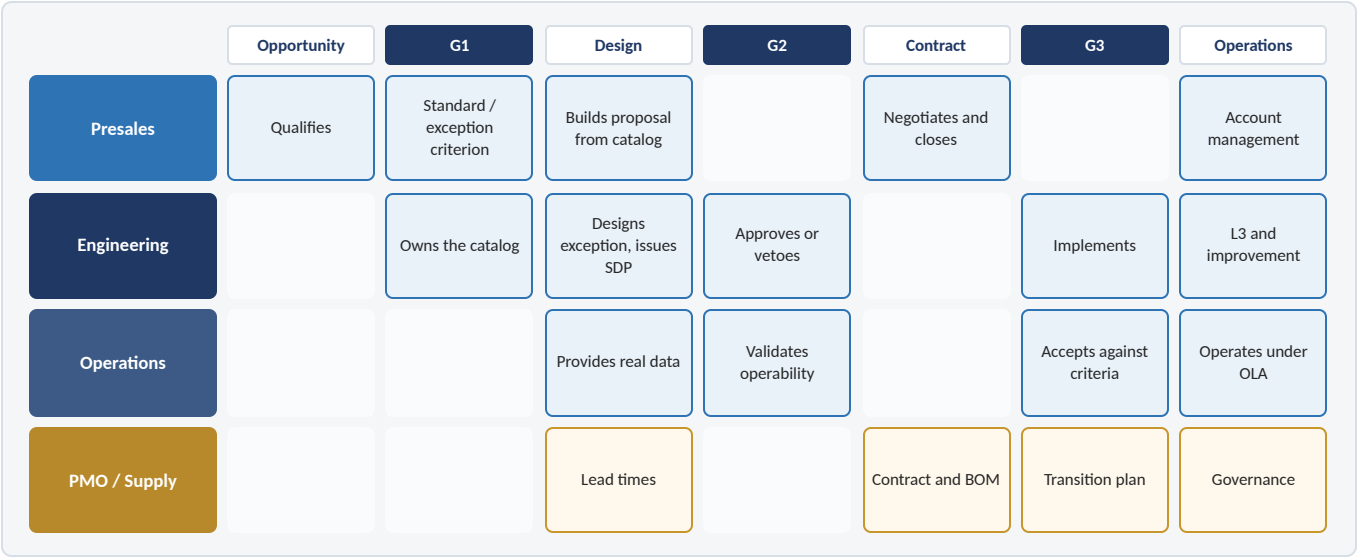
Solution design

What is sold to the client: topology, brands, scope, value proposition. A commercial activity, sensitive to time. It belongs to presales.

Service design

How the service is run and sustained for 24 to 36 months: levels, sizing, costs, profiles, tools, runbooks, acceptance and exit criteria. Governed by engineering.

THE FLOW WITH THREE GATES



Gate	What it decides	Who approves
G1 · Qualification	Whether the opportunity is standard or an exception. Off-catalog, new brand, different level or a service never operated goes to design.	Presales, with criteria defined by engineering
G2 · Operability	Whether operations can sustain what is about to be committed, at the proposed level and cost. Requires a joint signature.	Operations, with engineering review
G3 · Transition	Whether the service can go into production: tested runbooks, active monitoring, reconciled inventory, trained staff.	Operations accepts against criteria

THE MECHANISM THAT AVOIDS THE BOTTLENECK

If engineering reviewed every opportunity, no proposal would go out on time. The way out is to pre-design: a catalog of already approved service patterns —base level, capacity model, unit cost, profiles, tools, standard runbook— from which presales builds quotes by assembling blocks. The goal is for 80% of opportunities not to require an engineering cycle.

STANDARD PATH
80%
of opportunities solved from the catalog

PROPOSAL CYCLE
< 6 days
sustainable target with a published catalog

EXCEPTIONS
→ catalog
each exception sold once is industrialized

RESPONSIBILITY MATRIX

A: single approver · R: executes · C: consulted · I: informed

Activity	Presales	Architecture	Engineering	Operations	PMO
Opportunity qualification	A/R	C	I	I	I
Commercial proposal	A/R	C	I	I	C
Catalog and design standards	I	A	R	C	I
Service design package	C	A	R	C	I
Sizing and capacity	I	A	R	C	C
Operability validation	I	C	R	A	I
Final costing and lead times	C	I	C	I	A/R
Technical approval before signing	I	A/R	C	C	C
Implementation and transition	I	C	A/R	C	R
Operation under agreed levels	I	I	A	R	C
Design improvement	I	A	R	R	I

The two assignments that are always debated

In operability validation the approver is operations, not engineering: whoever is going to operate must be able to say no. Engineering keeps its own brake in the technical approval before signing. They are two different controls and neither replaces the other.

In daily operations, execution belongs to the vendor or operations, but accountability stays with delivery engineering, because the company answers to the client even when it does not execute.

WHAT I WOULD DO DIFFERENTLY

I would publish the catalog before activating the gates. If the governance model goes live without a catalog, every opportunity becomes an exception and engineering becomes exactly the bottleneck the design was meant to avoid. The order matters more than the design.

04 Executive operations dashboard

Seven indicators, three questions, one page

THE PROBLEM

Operational dashboards fail from excess, not from lack. Forty indicators on one screen produce the same paralysis as none. The discipline is to choose the smallest set that lets leadership answer three questions and decide.

Are we delivering?
Health of the committed service. If this fails, everything else is irrelevant.

Are we making money?
Real profitability of delivery, not the one projected at signing.

Can we sustain it?
Capacity, talent and future demand. The pillar nobody looks at until it breaks.

THE SEVEN INDICATORS

Pillar	Indicator	Why this one and not another
Service	Compliance with agreed levels	It is the contractual commitment: it defines whether there is a penalty
Service	First contact resolution	The only indicator that improves cost and experience at the same time
Service	Incident recurrence	Separates structural improvement from efficient firefighting
Finance	Real margin per contract	Against the designed margin: the deviation reveals the problem before the income statement does
Finance	Days sales outstanding	The service can be profitable while the account keeps destroying cash
Sustainability	Capacity utilization	Below 65% there is slack; above 85% it breaks
Sustainability	Team turnover	The most honest indicator of whether the operating model is sustainable

The design principle

Every indicator must be able to drill down to the evidence that produced it: from the percentage to the contract, from the contract to the ticket. A number that cannot be traced gets debated in the committee instead of being decided on. Traceability is not a dashboard feature, it is its condition for existing.

WHAT IT LOOKS LIKE (SYNTHETIC DATA)



THE READING THAT MATTERS

This synthetic dashboard tells a story that a service-only dashboard would hide: compliance improves for six months in a row, but margin deteriorates and turnover rises. It is the classic pattern of an operation buying performance with the team's effort. Service looks good today and will break within two quarters.

Why the sustainability pillar exists

Without it, leadership congratulates the team on the green indicator and finds out about the problem when three key people resign. Turnover and utilization are not HR metrics: they are the leading indicators of next year's service.

WHAT I WOULD DO DIFFERENTLY

I would add an eighth indicator for data quality —records without an assigned owner, closures without a root cause— in the first version, not the second. Every other number depends on it, and when it fails the dashboard is not wrong: it lies with confidence.

05 From commercial catalog to verifiable obligations

The matrix that translates what is sold into what must be delivered

THE PROBLEM

The commercial catalog says what each contract tier includes. Operations needs something else: who executes each service, who is accountable for it and what it delivers, how often and by when. There is usually a gap between those two documents, and that gap is where uncontracted work gets absorbed and margin is lost without anyone recording it.

TIER STRUCTURE (GENERIC FOUR-LEVEL EXAMPLE)

Element	Basic	Extended	Managed	Custom
Service window	8x5	24x7	24x7	Per contract
Mode	Reactive	Reactive	Reactive and managed	Proactive and managed
Monitoring	—	—	Included	Included
Root cause analysis	—	Included	Included	Included
Capacity management	—	—	Included	Per contract
Preventive maintenance	—	—	Included	—
Penalties	Do not apply	Do not apply	Apply	Apply

The observation that usually goes unnoticed

The extended tier has a continuous window with purely reactive service and no monitoring. That means sustaining 24-hour availability with no automatic detection: demand only comes in when the client reports. Its team is sized for coverage, not volume, and it is almost always the least profitable tier in the catalog. Spotting it before signing is the difference between a healthy contract and one you defend for two years.

EXCLUDED SERVICES, DECLARED AS SUCH

What is not included must be written with the same precision as what is. Without an explicit statement, operations ends up absorbing it in practice: on-call coverage for special events, management of the client's vendors, site surveys, work at heights, own stock of spare parts. Each one carries its operational implication, so the commercial conversation is an informed one.

THE COLUMN THAT TURNS A CATALOG INTO AN OPERATIONAL AGREEMENT

It is not the applicability column, but the execution column: who does each thing when the operation is split between the company and a third party.

Service	Practice	Execution and accountability
Service desk	Incidents	Operator executes · company answers to the client
Remote support L1 and L2	Incidents	Operator executes · escalates to own third level
Manufacturer follow-up	Incidents	Own supply manages · operator provides technical evidence
Normal change management	Changes	Operator executes · own engineering authorizes
Monitoring	Events	Vendor operates · thresholds and severities defined by the company
Capacity management	Capacity	Own engineering, with data delivered by the operator
Inventory and configuration	Assets	Operator maintains · the reference record belongs to the company
Continuous improvement	Improvement	Own engineering proposes · operator executes under change control

DELIVERABLES: THE COMMERCIAL ONES AND THE ONES NOBODY WRITES DOWN

Committed to the client

Deliverable	Deadline
On-site event report	24 business h
Root cause analysis	5 business days
Change control	Before executing
Monthly compliance report	Business day 3
Inventory report	Business day 5

Internal, not visible to the client

Deliverable	Frequency
Record-level operations data	Monthly
On-site inventory reconciliation	Quarterly
Staffing and certifications status	Quarterly
Design deviation log	Monthly
Grouped burst log	Monthly

Why the internal deliverables matter more

The commercial ones are reviewed in the committee with the client and nobody forgets them. The internal ones are claimed by no one, and they are precisely the ones that preserve the ability to design: without record-level operational data, the organization loses, within two or three contract cycles, the information it needs to size, quote and improve. It is a silent degradation.

WHAT I WOULD DO DIFFERENTLY

I would set the verification checklist before negotiating the first account, not after the third. Twelve points —confirmed window, distinction between response and resolution, request catalog, thresholds, penalty scheme, declared exclusions— prevent most of the discussions that later get resolved by conceding.

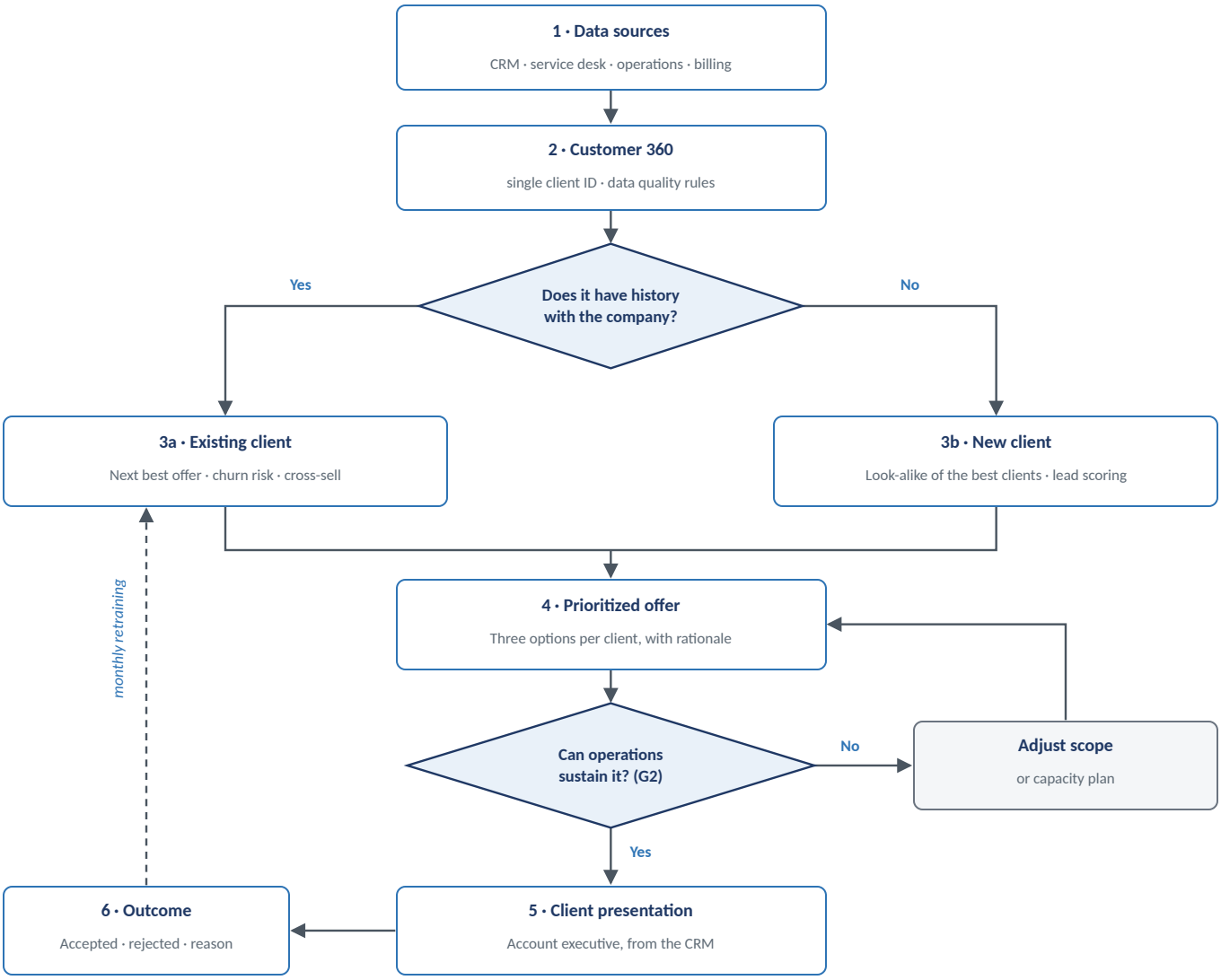
06 Commercial intelligence on operational data

What to offer each client, without selling what operations cannot sustain

THE PROBLEM

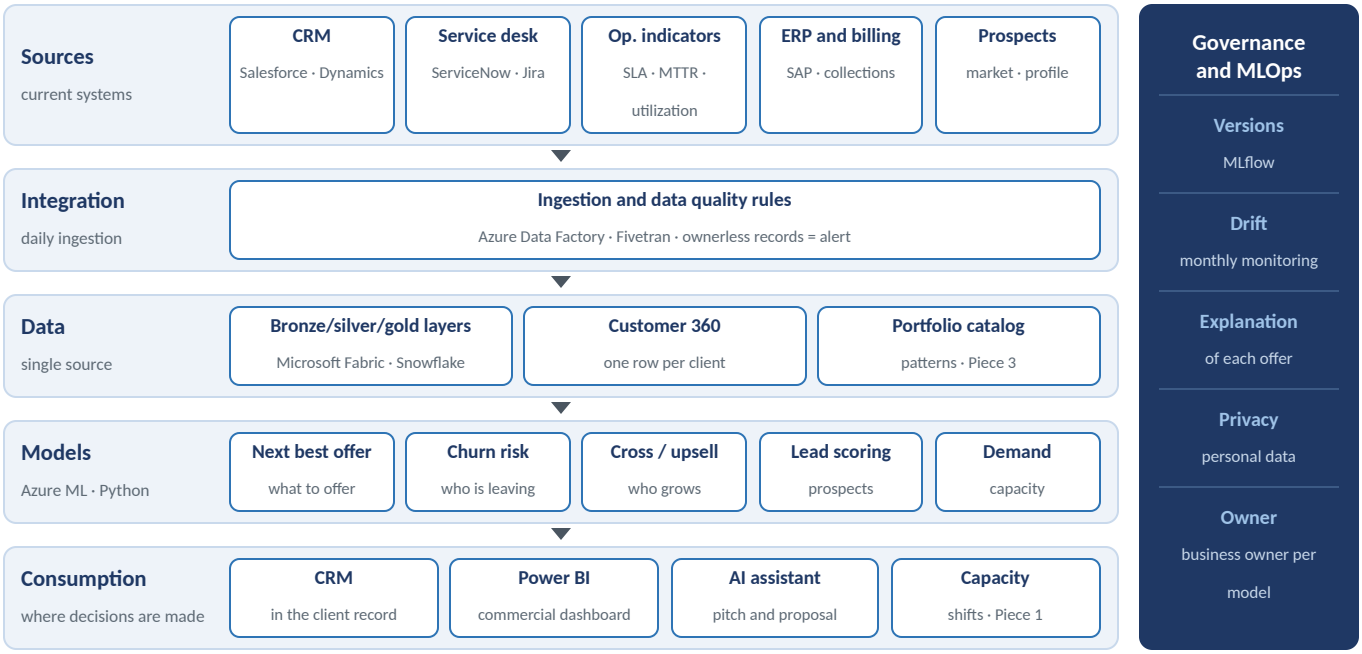
Sales decides what to offer by intuition and quarterly quota; operations knows which client is saturating its capacity, which one lives at the edge of its SLA and which one pays late. That information exists, but it lives in systems that do not talk to each other. The result: everyone gets the same offer, the client who was already sending signals is lost, and what is sold is what nobody can sustain later.

THE FLOW · FROM OPERATIONAL DATA TO THE OFFER



The lower diamond is the same operability gate from Piece 3: no recommendation reaches the client unless operations confirms it can sustain it at the proposed level and cost. Each outcome, accepted or rejected, retrains the model.

THE ARCHITECTURE · FIVE LAYERS AND CROSS-CUTTING GOVERNANCE



The tools are interchangeable examples: the architecture adapts to the ecosystem the company already has. What cannot be replaced is the Customer 360 layer and the governance.

FIVE MODELS, FIVE BUSINESS QUESTIONS

Model	Question	Signals used	Suggested technique
Next best offer	What should I offer this client now?	Purchase history, real service usage, similar clients	Gradient boosting · collaborative filtering
Churn risk	Who will leave if I do nothing?	Missed SLAs, recurrence, days outstanding, tickets	Classification · gradient boosting
Cross-sell and upsell	Who can grow, and toward which tier?	Contracted tier vs. real usage, exceeded capacity	Classification + catalog rules
Lead scoring	Which prospect looks like my best clients?	Industry, size, region, profile of the most profitable clients	Look-alike profiling
Demand forecast	Will I be able to sustain what is about to be sold?	Pipeline, seasonality, hourly demand curve	Time series

Why not a single neural network

With tabular business data —sales, tickets, indicators— gradient boosting models match or beat neural networks with less data, and they can be explained to a sales director. Neural networks are reserved for text: emails and tickets, where the reason for dissatisfaction is not in any structured field.

SYNTHETIC EXAMPLE · FOUR CLIENTS, FOUR DECISIONS

Client	Tier	Operational signal	Churn	Recommendation	Operable?
Client A	Basic 8x5	22% of tickets arrive outside business hours	Low	Extended with a wider window	Yes: shift redistribution, no hiring (Piece 1)
Client B	Managed	SLA 91% and 12% recurrence three months in a row	High	None: retention plan first	Not applicable until service recovers
Client C	Extended	Capacity at 92% for a whole quarter	Medium	Capacity management	Yes, within the catalog
Prospect D	—	Profile similar to the most profitable 10% of the portfolio	—	Managed	Requires operability validation (G2)

Client B is the case that justifies the whole model: without service data, sales would have offered an upgrade to a client who is about to leave.

THE OFFER · OPERATIONAL COMMERCIAL INTELLIGENCE IN THREE PHASES

Phase	Duration	What is delivered	Criterion to move forward
1 · Data diagnostic	4 weeks	Source inventory, single client ID, data quality report, prioritized use cases	80% of clients reconciled across CRM, service and billing
2 · Pilot	8–10 weeks	Churn risk and next best offer for one segment; recommendation visible in the CRM; dashboard	Executives use the recommendation in at least 30% of their accounts
3 · Scale-up	3–6 months	Five models, MLOps, link to capacity planning and a generative AI assistant	Real margin of what is sold equal to or above the designed margin

HOW IT IS MEASURED (ILLUSTRATIVE TARGETS)

CONVERSION
+15%
offers accepted vs. generic offer

RETENTION
−20%
churn in clients flagged at risk

PROPOSAL CYCLE
< 6 d
with pre-built offer and catalog

MARGIN
≥ 0 pp
real vs. designed deviation

THE FRAGILE ASSUMPTION

That the CRM, the service desk and billing share a client identifier. If they do not, the models learn from misattributed records and recommend with confidence on wrong data. That is why the first phase ends in client reconciliation, not in a model: it is the least glamorous work and the one that decides whether everything else is useful.

Where this offer fits

In operations leadership, managed services delivery or digital transformation roles, where operations stops being seen as a cost center and becomes a source of growth: the same data that sustains the SLA decides what to sell, to whom and when.

07 Patient 360: data and AI in a private hospital

Anticipating what each patient needs, and knowing what it costs to build and run

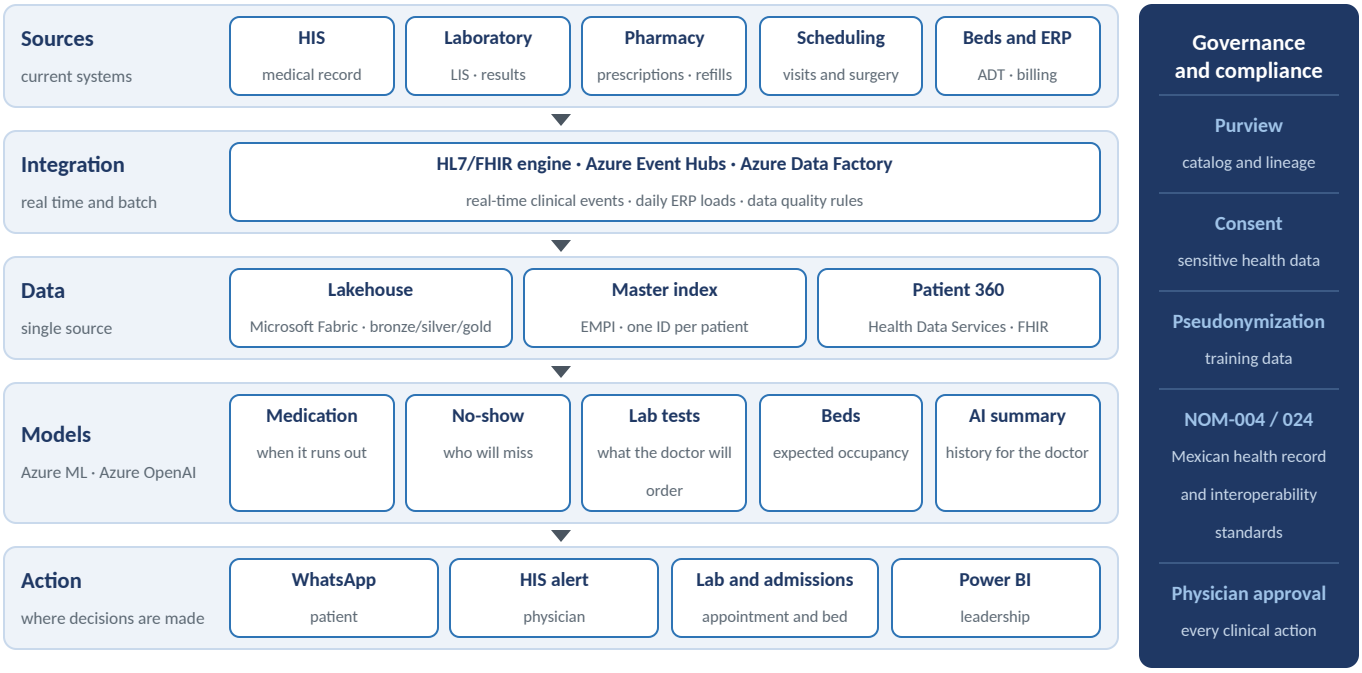
THE PROBLEM

A private hospital has each patient scattered across seven systems that do not talk to each other: medical records, scheduling, pharmacy, laboratory, operating rooms, beds and billing. The cost shows up as missed appointments, interrupted treatments, same-day surgery cancellations and balances discovered at discharge. The question is not which AI model to use, but how to turn seven partial records into one view that acts before the problem.

THE METHOD

1	2	3	4	5
Unify identity: a single patient ID (EMPI)	Build the 360 view on the FHIR standard	Models that anticipate, not that report	Orchestrate actions with physician approval	Measure the baseline and scale in phases

THE ARCHITECTURE · FIVE LAYERS AND CROSS-CUTTING GOVERNANCE



SIX MODELS, SIX BUSINESS QUESTIONS

Model	Question	Suggested technique
Medication depletion	When will it run out, and will the refill happen here?	Rules + adherence regression
No-show	Who is going to miss the appointment?	Classification · gradient boosting
Pre-visit lab tests	Which tests will the doctor order at the visit?	Clinical rules + recommendation
Bed demand	How many beds per day and type?	Time series
Readmission and collections	Which patient or which account is at risk?	Classification and scoring
Pre-visit summary	What does the doctor need to know in 30 seconds?	Generative AI with guardrails

THE FLOW · A SYNTHETIC PATIENT

María, 58, diabetic, with an endocrinology visit and a scheduled knee surgery.

Moment	What the system detects	Action
7 days before	Her metformin is running out	WhatsApp to refill; the doctor approves the prescription
10 days before	Upcoming visit without lab results	The lab offers her a slot for HbA1c
3 days before	High no-show risk	Reinforced reminder and option to reschedule
Visit day	Results ready	The doctor receives an AI-generated summary
15 days before	Surgery confirmed	Bed reservation and insurer validation
2 days before	A pre-op test is missing	Alert to the surgeon and the patient; same-day cancellation avoided

PLATFORMS, CAPEX AND OPEX PER INITIATIVE (USD)

Initiative	Platforms	Months	CAPEX	OPEX / month
1 · Data foundations	Fabric F64, EMPI, HL7 engine, Health Data Services, Purview	0–4	450K	14.5K
2 · Patient 360	Power BI (physician viewing included in F64)	2–5	120K	1.0K
3 · Automation	Logic Apps, Functions, WhatsApp Business	4–8	200K	5.5K
4 · Predictive models	Azure Machine Learning	8–12	250K	3.0K
5 · Generative AI	Azure OpenAI, AI Search, Content Safety	12–15	200K	3.5K
Change, PMO and team	Five-person operations team	—	100K	25.0K
Total		**15**	**1,320K**	**52.5K**

Equivalents: AWS (HealthLake, SageMaker, Bedrock) and GCP (Healthcare API, Vertex AI). Everything runs in the cloud; no hardware purchase.

THE RESULT (ILLUSTRATIVE BUSINESS CASE)

CAPEX
1.32 M
one-time implementation investment

OPEX
0.63 M
per year, platforms and team

BENEFIT
1.37 M
per year: no-shows, refills, surgeries and collections

PAYBACK
≈ 21 months
at full operation

The non-negotiable principle

AI proposes and the physician decides. No refill, test or reservation is executed without human approval, and controlled medications stay out of automation. In healthcare, without trust the most accurate model goes unused.

THE FRAGILE ASSUMPTION

That medical records, pharmacy and billing can be reconciled into a single patient identifier. If the master index fails, every model inherits duplicates and notifies the wrong person. It is the lesson from Piece 6: the first phase ends in identity, not in a model.

WHAT I WOULD DO DIFFERENTLY

I would put the first use case in the hands of the pharmacy, not technology. Refills are what patients and doctors notice first, and the lever that funds the next phases fastest.

08 Predictive operations in retail

Seeing the failure before the store does, and knowing what it costs

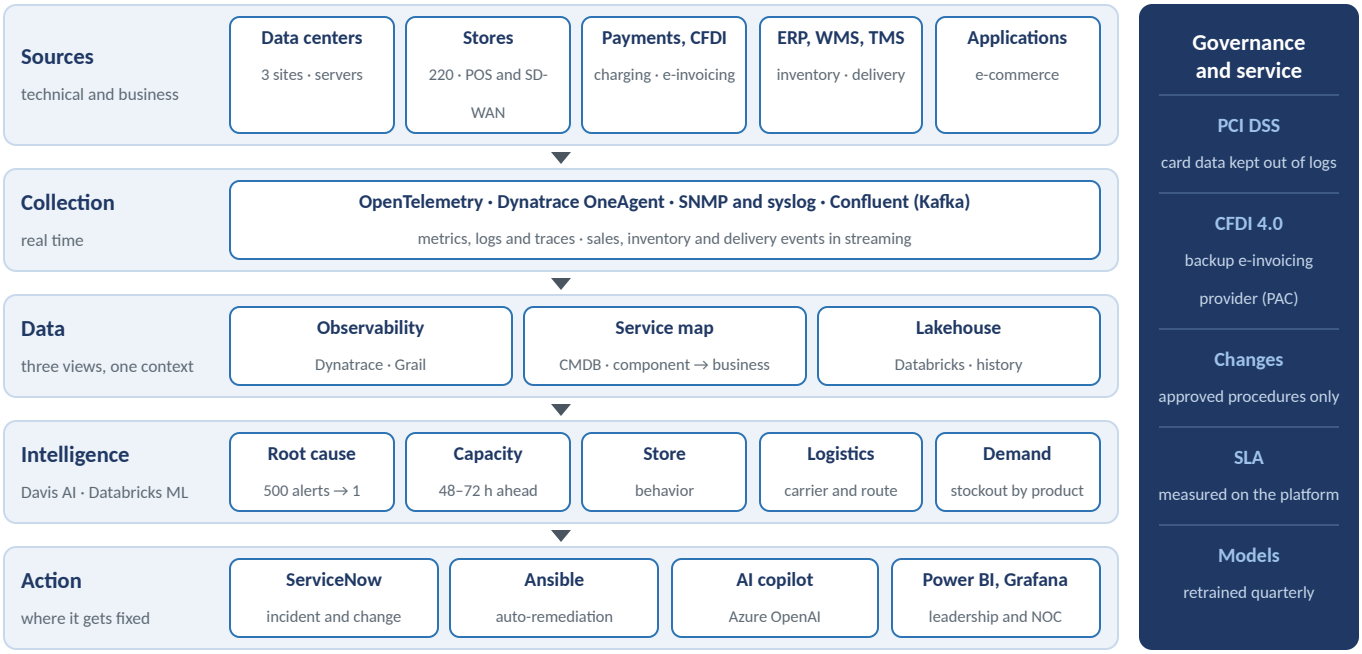
THE PROBLEM

A chain of more than 200 stores takes online and in-store payments on three data centers, under a support contract that includes the infrastructure. Today a failure is discovered when the register stops charging: thousands of uncorrelated alerts, no view of which sales or region are at risk, and inventory, deliveries and demand reviewed only after a stockout. Every outage costs sales, SLA penalties and trust.

THE METHOD

1. Instrument 3 data centers and 220 stores
2. Map every component to its line of business
3. Group alerts by root cause
4. Predict capacity and business deviations
5. Remediate only with approved procedures

THE ARCHITECTURE · FROM TECHNICAL SIGNAL TO SALES



SIX MODELS, SIX BUSINESS QUESTIONS

Model	Question	Suggested technique
Correlation and root cause	Which of the 500 alerts is the cause?	Topology graph and causality
Capacity saturation	When will payments or invoicing saturate?	Time series (Prophet, LSTM)
Log failures	Which pattern precedes an outage?	Log pattern clustering (Drain)
Store behavior	Which store sells or charges outside its pattern?	Isolation Forest, autoencoders
Carriers	Which carrier and which route are falling behind?	Delivery-time regression
Product demand	Which product will run out before replenishment?	Forecasting and anomaly detection

THE FLOW · A SYNTHETIC BUEN FIN (MEXICO'S BLACK FRIDAY)

Moment	What the system detects	Action
48 h before	Payments in the North DC will reach 92% capacity	Add capacity and balance with the West DC
Friday 18:40	Authorization latency 35% above its pattern	Davis AI groups 420 alerts into one incident
18:42	Root cause: a change in the payments database	Ansible rolls back the change with an approved procedure
18:44	Impact: 38 northern stores at risk	Notice to store managers and leadership
Saturday	The e-invoicing (CFDI) queue doubles	A second e-invoicing provider (PAC) is activated
Sunday	A carrier falls behind while demand rises	Replenishment from another distribution center

PLATFORMS, CAPEX AND OPEX PER INITIATIVE (USD)

Initiative	Platforms	Months	CAPEX	OPEX / month
1 · Observability	Dynatrace (DC, apps and logs), ThousandEyes (store network)	0–5	350K	52.0K
2 · AIOps and root cause	ServiceNow ITOM AIOps, CMDB and service map	3–8	300K	12.0K
3 · Business data	Databricks, Confluent (Kafka), Power BI	4–9	400K	20.0K
4 · Predictive models	Databricks ML	8–14	300K	5.0K
5 · Automation	Ansible Automation Platform, Azure OpenAI	12–18	250K	7.0K
Operations, PMO and team	Seven-person AIOps team	—	150K	35.0K
Total		**18**	**1,750K**	**131.0K**

Alternatives: Datadog or Splunk, BigPanda, Snowflake or Fabric, Rundeck. Assumes ServiceNow ITSM is already licensed. The three data centers stay on-premises; collectors run on the existing infrastructure.

THE RESULT (ILLUSTRATIVE 18-MONTH TARGETS)

MAJOR INCIDENTS
60 → 36
40% fewer per year

DETECTION
< 5 min
was 30 min; 70% before the store

RECOVERY
3 → 1.5 h
hours with sales impact: 180 → 54

PAYBACK
≈ 28 months
CAPEX 1.75 M · OPEX 1.57 M/yr · benefit 2.33 M/yr

The finding that changes the conversation

It is Piece 1 at scale: there, 17% of events were noise; here, one failure in the payments database produces hundreds of alerts. The value is not in reacting faster to each one, but in grouping them into one cause and linking it to the sales at risk. Without a service map, AI correlates but does not know what matters.

THE FRAGILE ASSUMPTION

The quality of the CMDB. Correlation and impact calculation depend on knowing which server supports which business service. With an outdated inventory, the model groups well and attributes badly; that is why phase 2 starts by reconciling inventory.

WHAT I WOULD DO DIFFERENTLY

I would negotiate retiring the current monitoring tools from phase 1. It is the saving the client feels first and the one that funds prediction, which is what takes longest to prove its value.

THE METHOD BEHIND THE EIGHT PIECES

How I work

The previous pieces deal with different problems, but they follow the same sequence. It is not a branded methodology: it is the discipline of not giving an opinion before measuring.

Five principles

1 · Raw data before the report

Aggregated reports hide exactly what needs to be seen. The hourly curve, the distribution by priority, the behavior of bursts: that only shows up in record-level detail. The data almost always already exists, and nobody has pulled it.

2 · Declare the fragile assumption

Every model rests on two or three assumptions, and usually one of them moves the result more than the others. Naming it explicitly makes the model defensible; hiding it makes it fragile. A model that does not declare where it can break is not a model, it is an opinion with a table.

3 · The artifact has to be usable

A diagnosis that ends in conclusions changes nothing. It ends in a template, a model with live formulas, a responsibility matrix, a shift proposal or a business case with CAPEX and OPEX: something someone can pick up on Monday and apply without translation.

4 · Adoption is part of the design, not a later phase

Transformation programs do not fail because of technical design, they fail because people do not adopt. If the model demands data entry nobody will do or governance nobody will sustain, it is badly designed even if it is correct. In AI this applies twice: the person who decides has to understand why the model recommends.

5 · Say what did not work

Every piece in this portfolio includes a section on what I would do differently. It is not modesty: it is the only section that shows independent judgment. Someone who can only narrate their successes has not yet learned from the project.

Where I add the most value

Situations where I fit

- Operations that grew without a method and do not know where they lose margin
- Transferring an operation to a third party while keeping control and accountability
- Multi-site or multi-country delivery that needs consistency without forced uniformity
- Data and AI programs that must prove return before scaling

What I usually find first

- The data exists and nobody has used it
- The team is not wrongly sized, it is wrongly distributed
- What is not declared as an exclusion gets absorbed in practice
- AI is requested before the data is reconciled
- The green indicator hides an exhausted team

CONVERSATIONS I AM INTERESTED IN

Operations that must sustain more with the same

Sizing and capacity with real data

Service governance and outsourced operations

Multi-site and multi-country operating models

Analytics and automation applied to delivery

Commercial intelligence with AI on operational data

Data and AI programs with a business case, CAPEX and OPEX

AIOps and predictive operations on critical infrastructure

Lázaro Gómez Ontiveros

lazarogomez@hotmail.com

+52 55 7980 3050

linkedin.com/in/lazarogomezontiveros

Mexico City · available for travel and relocation · Spanish, English and Portuguese

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